

Project Assignment Report: Traffic Signal Synchronization System

Project Title: Traffic Signal Synchronization System Using Chinese Remainder Theorem
Course / Assignment: Number Theory & Discrete Mathematics Application
Team Leadership: Justin Joe Mathew (Team Lead)

Does Traffic Signal Synchronization Demo CRT or just LCM/GCD?

Mathematical Distinction:

1. LCM (Least Common Multiple) / GCD:

Used when all signals start at Minute 0 with NO remainder/offset. If Signal 1 (3 min), Signal 2 (5 min), and Signal 3 (7 min) all turn green at Minute 0, the next simultaneous green light is simply $\text{LCM}(3, 5, 7) = 105$ minutes.

2. CRT (Chinese Remainder Theorem):

REQUIRED when signals have DIFFERENT REMAINDERS / OFFSETS.

In our project scenario:

- Signal 1 (3-min cycle) turned green 2 minutes ago $\rightarrow x \equiv 2 \pmod 3$
- Signal 2 (5-min cycle) turned green 3 minutes ago $\rightarrow x \equiv 3 \pmod 5$
- Signal 3 (7-min cycle) turned green 2 minutes ago $\rightarrow x \equiv 2 \pmod 7$

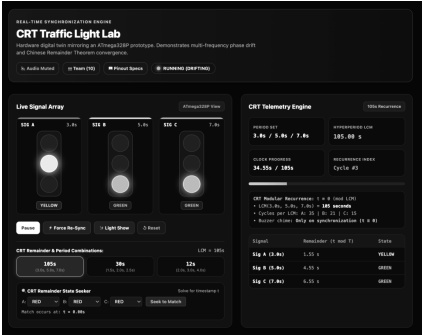
Because the remainders (2, 3, 2) are non-zero and distinct, LCM cannot solve this problem. The system requires the Chinese Remainder Theorem (Extended Euclidean Algorithm & Modular Inverses) to solve for the unique green wave minute ($x = 23$).

Conclusion: The project is a genuine demonstration of CRT, specifically showcasing why CRT is needed over simple LCM when real-world cycles start at different offsets.

Physical Hardware Prototype & Digital Twin Interface

The project includes both a physical Arduino Nano breadboard circuit and a full digital twin web simulator.

Interactive Web App & Project Repository: <https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem>
(Local Dev Server: <http://localhost:5173/>)



Multi-Channel Phase Waveform

Waveform visualization showing three channels (SIN A, SIN B, SIN C) plotted against time. The plot area is dark with light-colored waveforms. The x-axis is labeled 'Time' and the y-axis is labeled 'Amplitude'.

Arduino Nano Physical Firmware

arduino/traffic_light_controller.ino

Signal A: Pin 2 (Red), Pin 8 (Yellow), Pin 4 (Green) - Signal B: Pin 8 (Red), Pin 9 (Yellow), Pin 10 (Green) - Signal C: Pin A0 (Red), Pin A1 (Yellow), Pin A2 (Green) - Buzzer: Pin A1

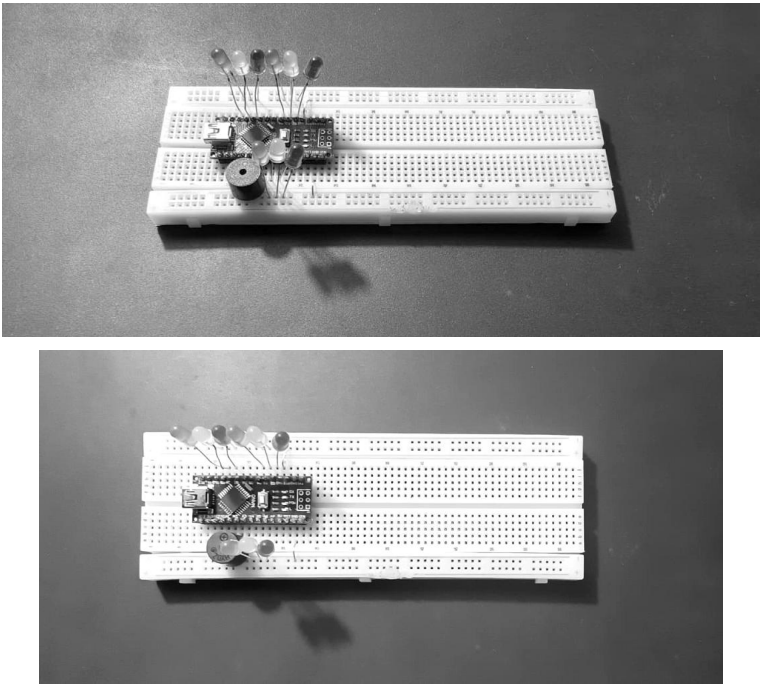
const int RED_A = 2;
const int YELLOW_A = 8;
const int GREEN_A = 4;

const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 10;

const int GREEN_C = A2;
const int YELLOW_C = A1;
const int RED_C = A0;

const int BUZZER = A1;

const unsigned long PULSE_A = 1000;
const unsigned long PULSE_B = 1000;
const unsigned long PULSE_C = 1000;
const unsigned long PULSE_CYCLE = 10000;



■ Group Member Roles & Work Division (10 Members)

PROJECT TEAM STRUCTURE		
■ 1. Project Leadership & Model	■ Justin Joe Mathew (Team Lead) & Jyothis Liju	■
■ 2. Hardware Sourcing & Build Testing	■ Joseph Alex & Joseph J	■
■ 3. Web Design & Frontend	■ Johan Geo & Johaan Sam	■
■ 4. Presentation & Showcase	■ Joel Geo Manuel	■
■ 5. Mathematics & Applications	■ Joel Duke, Jyothika Prakash, & Jose Alex	■

■ Detailed Team Member Contribution Breakdown

1. Project Leadership & Working Model Construction

■ Team Members: Justin Joe Mathew (Team Lead) & Jyothis Liju

- Justin Joe Mathew (Team Lead): Organized team work, built the physical 3-light breadboard model, and wrote the Arduino C++ code (`traffic\_light\_controller.ino`) for signal timing and buzzer sounds.
- Jyothis Liju: Co-developed the breadboard circuit model, plugging in LEDs to digital pins 2–10, connecting the buzzer to pin A1, and testing hardware timing against the written code.

2. Hardware Sourcing & Application Research

■ Team Members: Joseph Alex & Joseph J

- Joseph Alex: Bought the Arduino Nano board, breadboards, 220Ω resistors, LEDs, and buzzer, tested circuit connections, and checked pin voltage outputs.
- Joseph J: Handled component acquisition, tested circuit wiring, and researched real-world uses like city traffic light timing, fiber-optic channels, and computer processors.

3. Web Design & Web App Development

■ Team Members: Johan Geo & Johaan Sam

- Johan Geo: Designed the dark theme visual layout, circular LED monitors, interactive dials, and responsive dashboard cards.
- Johaan Sam: Built the website logic using React and TypeScript, coded the animation timer, remainder solver, buzzer sound triggers, and cross-device testing.

4. Presentation & Demo Walkthrough

■ Team Member: Joel Geo Manuel

- Joel Geo Manuel: Wrote the 2-speaker presentation script between Alex and Sam over team practice sessions, designed the demo slide layout, and timed the acoustic buzzer chime.

5. Mathematics & Educational Content

■ Team Members: Joel Duke, Jose Alex, & Jyothika Prakash

- Joel Duke: Worked out the 3s, 5s, and 7s cycle equations, coprimality proofs, and modulo inverse calculations ($y_1=2, y_2=1, y_3=1$).
- Jose Alex: Double-checked all modulo inverse calculations by hand, verified solution bounds ($M=105$), and checked edge cases for accuracy.
- Jyothika Prakash: Created the 3-step hint system on the website and wrote clear, simple guides explaining remainder modulo concepts without complex textbook jargon.